

**DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND
DEVELOPMENT**

INVENTORY & WAREHOUSE MANAGEMENT SYSTEM

Section - 1

**2520030601
2520030082
2520030184**

**Billakanti Siri
Bhavana Kondakrindi
Aniketh Pani**

INTRODUCTION

Inventory management is essential for businesses handling products across multiple warehouses.

Our system provides a centralized platform to manage:

- **Manage products & stock**
- **Track multiple warehouses**
- **Monitor stock movement**
- **Improve inventory visibility**

The system aims to make inventory management more accurate, organized, and efficient.



ABSTRACT

The Inventory & Warehouse Management System is a database-driven web application designed to streamline inventory operations across multiple warehouses. A MySQL relational database maintains structured relationships between products, warehouses, suppliers, purchase orders, and stock movements.

It incorporates reorder-level automation to identify low-stock products and support timely replenishment.

The system aims to reduce manual errors, improve stock visibility, and support efficient inventory decisions through centralized data management and reporting.

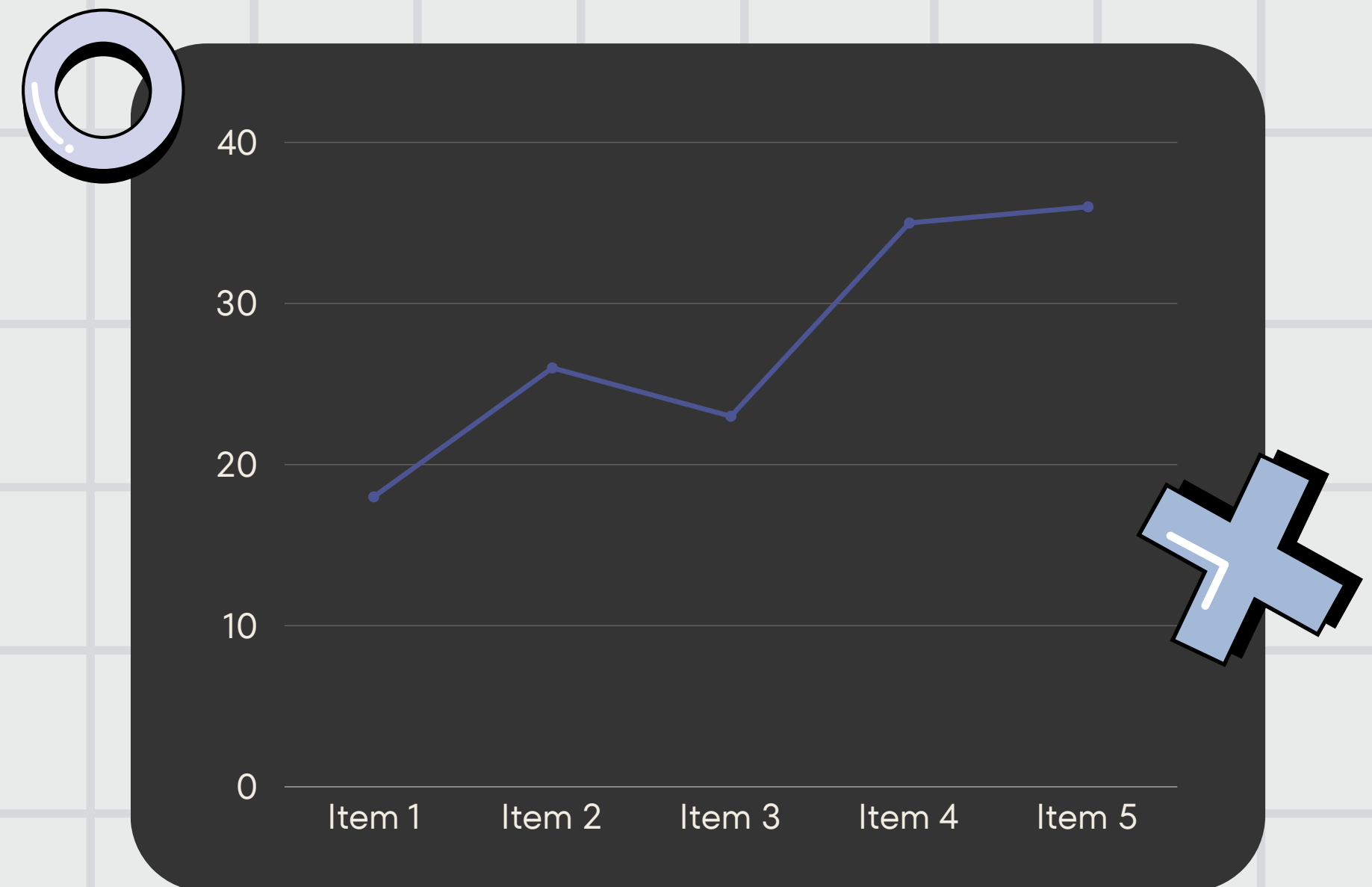
PROBLEM STATEMENT

EXISTING CHALLENGES

- Manual inventory records can lead to stock inaccuracies and human errors.
- Disconnected records make multi-warehouse tracking difficult.
- Stock transfers and movements may be difficult to monitor.

OUR APPROACH

A centralized database-driven system to improve stock accuracy, visibility, and warehouse operations.



MODULES

Warehouse & Inventory Management

- Track stock levels and product availability across multiple warehouses.
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Stock In / Stock Out

- Record incoming and outgoing inventory with accurate stock updates.
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Warehouse Transfer

- Transfer products between warehouses while maintaining updated stock records.
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Supplier & Purchase Orders

- Manage supplier information and create and track purchase orders.
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Reorder Level Automation

Identify products below their defined reorder level and support timely replenishment.

Stock Audit & Reports

Monitor stock activity and generate reports for inventory analysis and decision-making.

HARDWARE REQUIREMENTS

- **Processor: Intel i3 / equivalent or higher**
- **Memory: 4 GB RAM or higher**
- **Storage: 10 GB available space**
- **Display: 1366 × 768 or higher**

SOFTWARE REQUIREMENTS

- **Frontend - React.js**
- **Backend - Node.js + Express.js**
- **Database - MySQL**
- **Visualization - Chart.js**
- **Platform - Windows / macOS / Linux**

CONCLUSION

The Inventory & Warehouse Management System provides a centralized platform for managing inventory across multiple warehouses.

Key Outcomes

- **Accurate stock tracking**
- **Efficient warehouse operations**
- **Timely inventory replenishment**
- **Centralized supplier & purchase management**
- **Inventory monitoring and dashboard insights**

THANK YOU